

Delivery Logistics Performance & Delay Analytics

Data Analytics with AI

BharatCares Internship Project

Student Project Report

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Abstract

This report presents the findings of a data analytics project conducted as part of the BharatCares Data Analytics with AI internship programme. The project analyses a synthetic delivery-logistics dataset of 25,000 delivery records drawn from a simulated Indian multi-partner logistics environment. The objective is to understand patterns in delivery delays and to examine how delay rates vary across delivery modes, weather conditions, distance bands, delivery partners, and geographic regions. The analysis also investigates the relationship between delivery time and customer satisfaction ratings.

The analytical workflow is implemented entirely in Python using Pandas, NumPy, and Matplotlib. It covers data loading, data inspection, data cleaning, exploratory data analysis, business KPI calculation, dimensional analysis, and visualization. Key findings include an overall delay rate of 26.68%, a substantial variation in observed delay rates across delivery modes, a positive association between distance and both delivery cost and delay rate, and a weak negative correlation between delivery time and customer rating.

Note: The dataset used in this project is synthetic. All findings are observations from this dataset only and do not represent real-world performance of any logistics company.

1. Introduction

Timely delivery is one of the most important indicators of performance in the logistics and e-commerce industry. Delays in delivery directly affect customer satisfaction, operational costs, and the reputation of logistics providers. As the volume of deliveries grows — driven by the expansion of online shopping and same-day delivery services — the ability to understand and monitor delivery performance becomes increasingly important.

Data analytics provides a structured approach to examining large volumes of delivery records and identifying patterns that may not be visible through individual inspection. By grouping data across dimensions such as delivery mode, weather conditions, distance, and geographic region, analysts can observe which factors are associated with higher or lower delay rates and where performance differs across operational segments.

This project applies a Python-based analytics workflow to a synthetic delivery-logistics dataset. The goal is to demonstrate a complete analytical pipeline — from raw data cleaning to business insight generation — and to produce visualizations and structured findings that could support operational monitoring in a logistics context.

2. Problem Statement

The central analytical question addressed by this project is:

What patterns exist in delivery delays across delivery modes, weather conditions, distance bands, delivery partners, and regions in this synthetic logistics dataset? How are delivery time and customer satisfaction ratings associated in this data?

The project does not attempt to prove that any single factor causes delivery delays. All findings are described as observed associations within this synthetic dataset.

3. Objectives

- Measure overall delivery performance across all 25,000 records.
- Calculate the overall delivery delay rate using the original delayed field.
- Analyse delay rates by delivery mode.
- Examine the association between weather conditions and observed delays.
- Analyse the relationship between delivery distance, delivery time, and cost.
- Examine the association between delivery time and customer ratings.
- Analyse delivery partner and regional delay-rate patterns.
- Visualize key findings through nine clearly labelled charts.
- Produce a reproducible, beginner-readable Python-based analytics workflow.

4. Dataset Description

The dataset used in this project is a synthetic delivery-logistics dataset containing 25,000 records and 15 columns. It simulates delivery operations across multiple Indian logistics partners, regions, delivery modes, and weather conditions. No personal information is contained in the dataset.

Table 1 describes the original columns of the dataset.

Column	Description
delivery_id	Delivery identifier (not a unique key in this dataset)
delivery_partner	Name of the logistics company handling the delivery
package_type	Category of the package (e.g. groceries, electronics, clothing)
vehicle_type	Vehicle used (e.g. bike, truck, ev van, scooter)
delivery_mode	Service tier: same day, express, two day, or standard
region	Geographic region: north, south, east, west, or central
weather_condition	Weather at delivery time: clear, cold, foggy, hot, rainy, or stormy
distance_km	Delivery distance in kilometres (range: 3.6 – 297.1 km)
package_weight_kg	Package weight in kilograms (range: 0.67 – 49.52 kg)
delivery_time_hours	Actual delivery time (stored as an encoded timestamp string)
expected_time_hours	Expected/SLA delivery time (stored as an encoded timestamp string)
delayed	Original delay label assigned to the record: yes or no
delivery_status	Final delivery outcome: delivered, delayed, or failed
delivery_rating	Customer satisfaction rating: 1 (lowest) to 5 (highest)
delivery_cost	Cost of the delivery in Indian Rupees

Table 1: Original dataset columns and their descriptions.

During the data-cleaning step, six additional derived columns were added to the working dataset: `delivery_time_hours_num` (numeric hours), `expected_time_hours_num` (numeric hours), `zero_hour_flag`, `is_delayed_flag`, `delayed_original_binary`, and `computed_delayed`. The original raw file was never modified.

5. Tools and Technologies

Tool / Library	Version	Purpose
Python	3.8+	Core programming language
Pandas	Latest	Data loading, cleaning, grouping, and analysis
NumPy	Latest	Numeric operations and NaN handling
Matplotlib	Latest	Chart and visualization generation

Table 2: Technologies used. All packages are listed in `requirements.txt`.

6. Methodology

The project follows a structured analytics pipeline implemented in a single Python script (`final_project.py`). The pipeline covers seven stages:

6.1 Data Loading

The raw CSV file (`Delivery_Logistics.csv`) is loaded into a Pandas DataFrame using `pd.read_csv()`. The file is opened in read-only mode and is never overwritten.

6.2 Data Inspection

The script inspects dataset shape, column data types, missing values, duplicate `delivery_id` values, categorical value distributions, and numerical field summaries. This step establishes data quality awareness before any analysis is performed.

6.3 Data Cleaning

Two fields (`delivery_time_hours` and `expected_time_hours`) are stored as encoded timestamp-like strings in the raw file. A custom parsing function extracts the nanosecond integer from each string to recover the hour value. 255 records where actual delivery time was zero are flagged (`zero_hour_flag = 1`) and treated as missing actual delivery time (NaN) in the cleaned working data. The original `delayed` column is preserved unchanged. Duplicate `delivery_id` rows are identified but not removed — all 25,000 records are retained.

6.4 Exploratory Data Analysis

The cleaned data is analysed for distributions across delivery modes, weather conditions, regions, partners, vehicle types, and package types. Pearson correlation is computed across numerical columns. SLA-tier delay rates and equal-time conflict counts are reported as data-quality observations.

6.5 Business KPI Analysis

Six overall KPIs are calculated: total deliveries, delay rate, average actual delivery time, average expected/SLA time, average delivery cost, and average rating. The original delayed field is used as the source of truth for all delay-rate calculations throughout the project.

6.6 Visualization

Nine charts are generated using Matplotlib and saved as PNG files to the `business_analysis_outputs/` folder. Charts cover KPI summary, delay by mode, delay by weather, distance vs time and cost, delivery time vs rating, status distribution, and partner/region delay comparisons.

6.7 Findings Generation

A structured findings report (`findings.json`) is generated, containing metric values, group-level comparisons, plain-language interpretations, and analytical cautions for each section.

7. Exploratory Data Analysis

After cleaning, the working dataset has 25,000 rows and 21 columns. The raw dataset has 25,000 rows and 15 columns.

Key EDA findings:

- Missing values: 255 NaN values in `delivery_time_hours_num` (the 255 zero-hour records treated as missing). All other columns are complete.
- Duplicate `delivery_id`: 500 records are associated with 2 `delivery_id` values that appear multiple times. These are preserved as data-quality observations; no records are removed.
- Delivery modes: 4 modes (express: 6,233 | same day: 6,279 | standard: 6,186 | two day: 6,302).
- Weather conditions: 6 conditions, each with approximately 4,100–4,220 records.
- Regions: 5 regions (east: 4,919 | north: 4,949 | south: 4,977 | west: 5,095 | central: 5,060).
- Delivery partners: 9 partners, each with 2,700–2,830 records.
- Delivery ratings: integer scale 1–5; average 3.67.
- Distance: 3.6 – 297.1 km; average $\approx$ 150 km.
- Delivery cost: ₹95.67 – ₹1,632.72; average $\approx$ ₹864.94.
- Delivery status: delivered 18,331 | delayed 5,341 | failed 1,328.
- Original delayed field: yes: 6,669 | no: 18,331.

- Data quality note: comparing the original delayed field with a strict 'actual time > expected time' rule shows 1,203 records where the two definitions disagree. The original delayed field is used for all business delay-rate calculations.

8. Key Performance Indicators

Table 3 presents the six overall business KPIs calculated from the dataset. All values are computed from the data; delay rate uses the original delayed field.

KPI	Value
Total deliveries	25,000
Overall delay rate (delayed = yes)	26.68%
Delivered (status)	18,331 (73.3%)
Delayed (status)	5,341 (21.4%)
Failed (status)	1,328 (5.3%)
Average actual delivery time	6.31 hours
Average expected / SLA time	13.11 hours
Average delivery cost	₹864.94
Average delivery rating	3.67 / 5

Table 3: Overall KPIs (source: *kpis.json*).

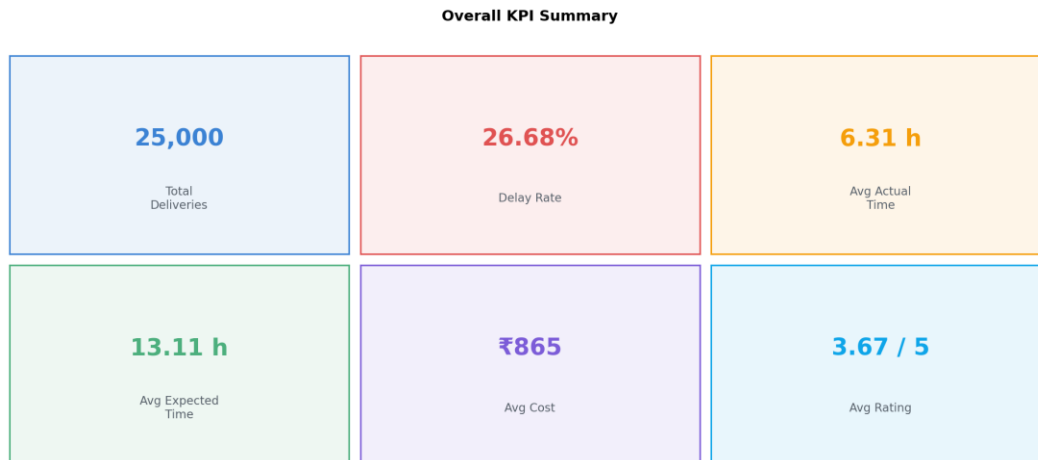


Figure 1: Overall Delivery KPI Summary.

9. Delivery Mode Analysis

Table 4 shows the delay rate and average delivery times for each of the four delivery modes.

Mode	Total	Delayed	Delay Rate	Avg Actual Time	Avg SLA Time
Express	6,233	4,599	73.78%	6.28 h	4.52 h
Same Day	6,279	2,043	32.54%	6.33 h	8.00 h
Two Day	6,302	27	0.43%	6.34 h	16.00 h
Standard	6,186	0	0.00%	6.30 h	24.00 h

Table 4: Delivery mode analysis (source: mode_analysis.csv).

The analysis observes that express mode is associated with the highest delay rate in this dataset (73.78%), while standard mode shows 0%. The wide spread of 73.78 percentage points across modes is a notable pattern. However, delivery modes differ in their SLA windows — express mode has a tighter SLA (average 4.52 hours) compared to standard mode (24 hours). This difference in SLA threshold likely contributes to the higher observed delay rate for express deliveries in this dataset. This is an association; the analysis does not conclude that the mode itself causes delays.

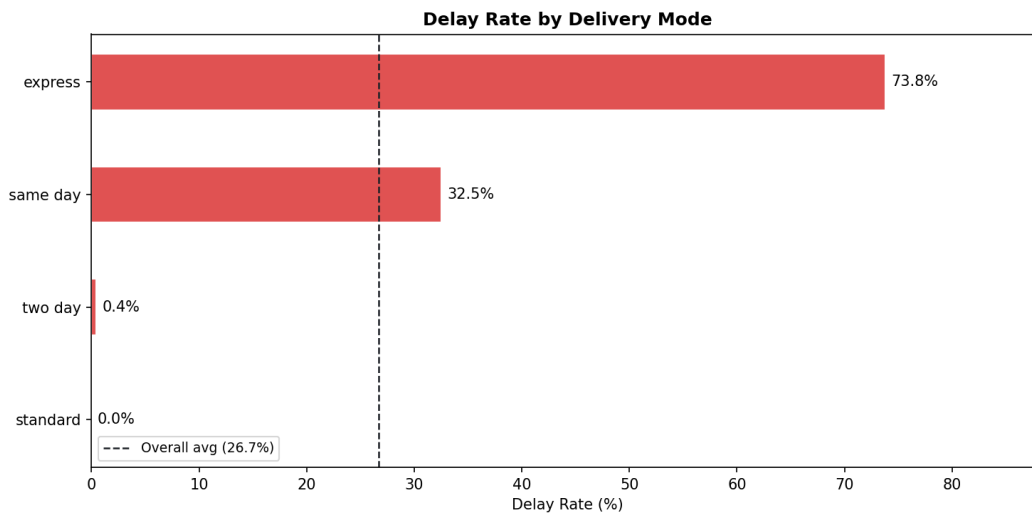


Figure 2: Delay Rate by Delivery Mode.

10. Weather Analysis

Table 5 presents delay rates and average delivery times across six weather conditions.

Condition	Total	Delayed	Delay Rate	Avg Time
Stormy	4,198	1,740	41.45%	8.81 h
Rainy	4,171	1,558	37.35%	7.80 h
Foggy	4,219	1,279	30.32%	6.49 h
Clear	4,124	719	17.43%	4.89 h
Hot	4,130	707	17.12%	4.87 h

Cold	4,158	666	16.02%	4.87 h
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Table 5: Weather condition analysis (source: *weather_analysis.csv*).

Adverse weather conditions — stormy (41.45%), rainy (37.35%), and foggy (30.32%) — are associated with higher delay rates and longer average delivery times in this dataset. Milder conditions (clear, hot, cold) show delay rates in the range of 16–17%. The spread across conditions is 25.43 percentage points. This is an observed association in synthetic data and does not establish a causal relationship.

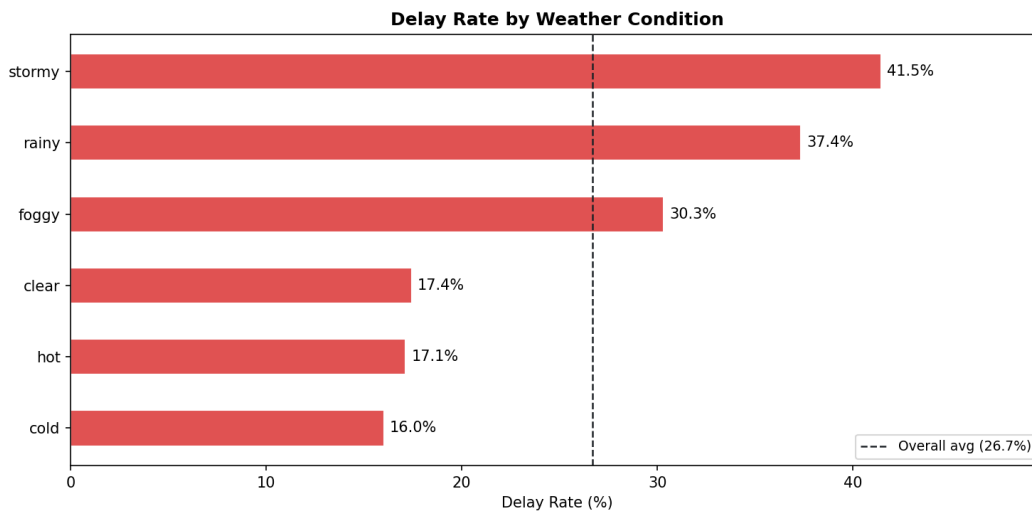


Figure 3: Delay Rate by Weather Condition.

11. Distance Analysis

Table 6 presents delay rate, average delivery time, and average cost across five distance bands.

Distance Band	Count	Delay Rate	Avg Time	Avg Cost
0–60 km	4,920	17.36%	3.42 h	₹264.36
61–120 km	5,044	19.85%	4.74 h	₹560.62
121–180 km	5,053	26.04%	6.29 h	₹866.35
181–240 km	4,960	31.75%	7.69 h	₹1,162.78
241–300 km	5,023	38.28%	9.25 h	₹1,463.31

Table 6: Distance band analysis (source: distance_analysis.csv).

Both average delivery time and average delivery cost increase consistently with distance in this dataset. Delay rate also increases across distance bands, from 17.36% for the 0–60 km band to 38.28% for the 241–300 km band. The cost rises from ₹264.36 to ₹1,463.31 across the same bands. These are clear positive associations between distance and each of these metrics. Distance alone does not explain all delay variation, as other variables (mode, weather) are not controlled for in this analysis.

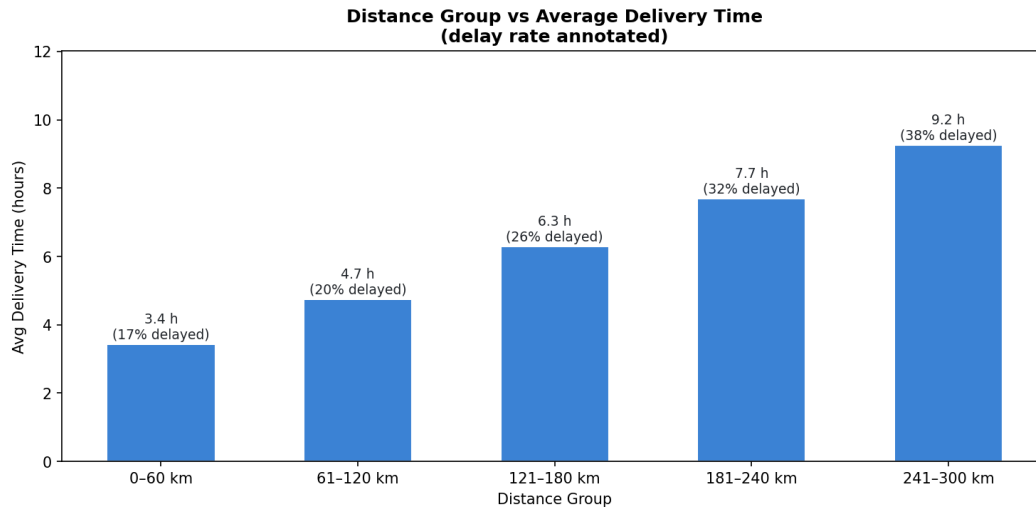


Figure 4: Distance Group vs Average Delivery Time (delay rate annotated).

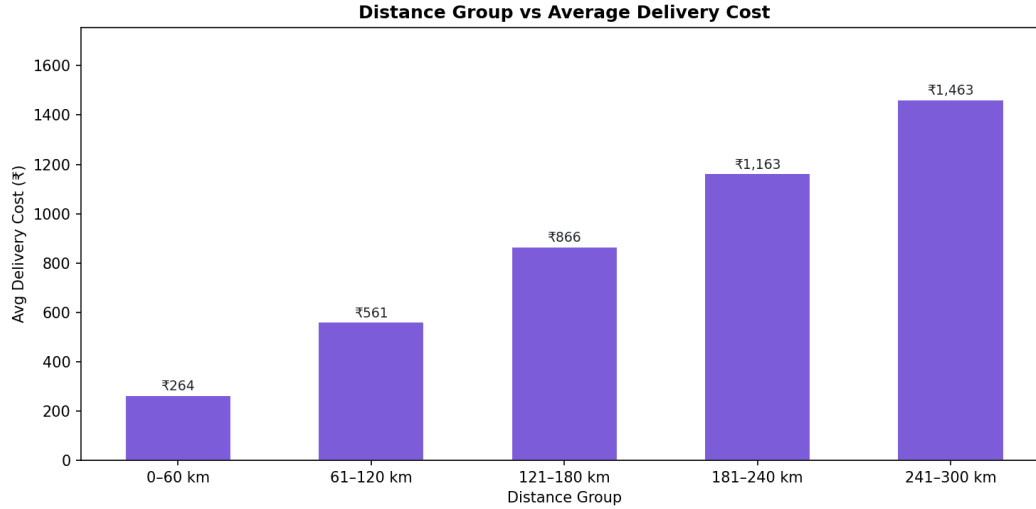


Figure 5: Distance Group vs Average Delivery Cost.

12. Customer Experience Analysis

The Pearson correlation between actual delivery time and customer rating is $r = -0.2693$, indicating a weak negative association. Table 7 shows average delivery time, average cost, and delay rate for each rating level.

Rating	Count	Avg Delivery Time	Avg Cost	Delay Rate
1 (lowest)	920	8.13 h	₹993.76	100.00%
2	3,602	8.11 h	₹1,023.27	100.00%
3	5,773	6.53 h	₹886.64	37.19%
4	7,318	5.64 h	₹807.18	0.00%
5 (highest)	7,387	5.69 h	₹811.97	0.00%

Table 7: Delivery rating analysis (source: rating_analysis.csv).

Ratings 1 and 2 correspond to 100% delay rates in this dataset, while ratings 4 and 5 show 0% delay rates. This strong alignment between delay status and rating is a pattern in the data. The Pearson $r = -0.2693$ represents a weak negative association across all records. Correlation does not establish causation — other factors such as package condition, communication quality, and delivery cost also influence customer satisfaction but are not fully captured in this dataset.

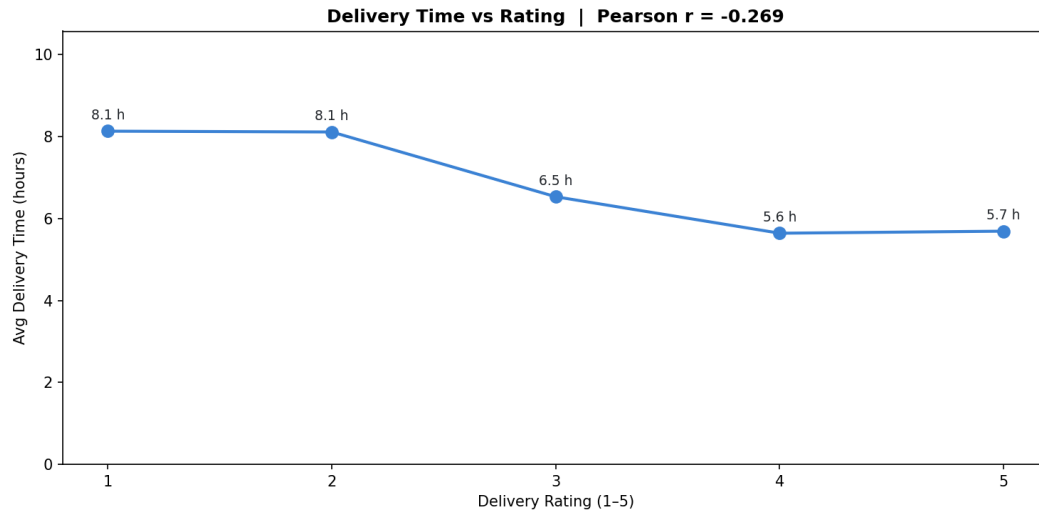


Figure 6: Delivery Time vs Customer Rating (Pearson $r = -0.2693$).

13. Delivery Status Analysis

The dataset records three delivery outcomes: delivered, delayed, and failed. Table 8 shows the distribution.

Status	Count	Percentage
Delivered	18,331	73.3%
Delayed	5,341	21.4%
Failed	1,328	5.3%
Total	25,000	100.0%

Table 8: Delivery status distribution.

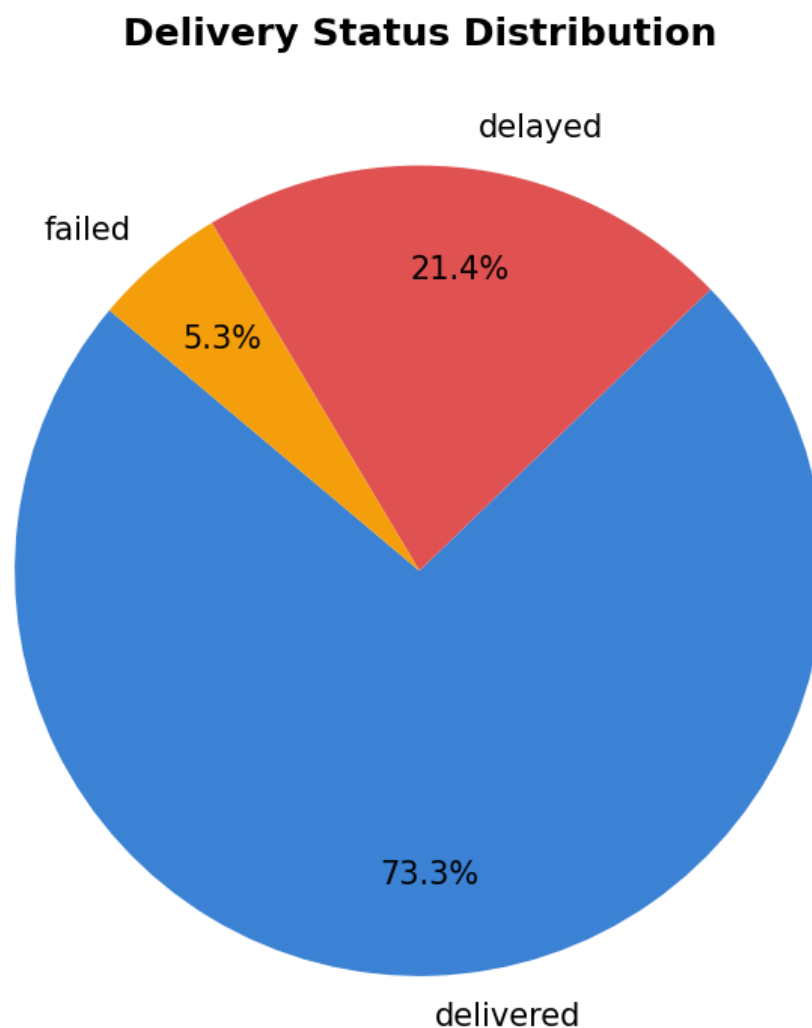


Figure 7: Delivery Status Distribution (Pie Chart).

14. Delivery Partner Analysis

Table 9 presents the observed delay rate and average customer rating for each of the nine delivery partners. Partners are listed in ascending order of delay rate.

Partner	Total	Delayed	Delay Rate	Avg Rating
Delhivery	2,786	691	24.80%	3.69
FedEx	2,818	709	25.16%	3.70
DHL	2,802	738	26.34%	3.69
Ecom Express	2,722	725	26.63%	3.67
Blue Dart	2,798	754	26.95%	3.68
Amazon Logistics	2,711	738	27.22%	3.64
Shadowfax	2,736	747	27.30%	3.66
Ekart	2,801	768	27.42%	3.66
Xpressbees	2,826	799	28.27%	3.61

Table 9: Delivery partner analysis (source: partner_analysis.csv). Sorted ascending by delay rate. No ranking is implied.

The delay rates across the nine partners range from 24.80% to 28.27%, a spread of 3.47 percentage points. This is a relatively narrow range in this synthetic dataset. Because partners may serve different regions, carry different package types, or operate across different delivery modes, the observed differences reflect a descriptive pattern rather than an isolated measure of each partner's performance. No partner is ranked as best or worst.

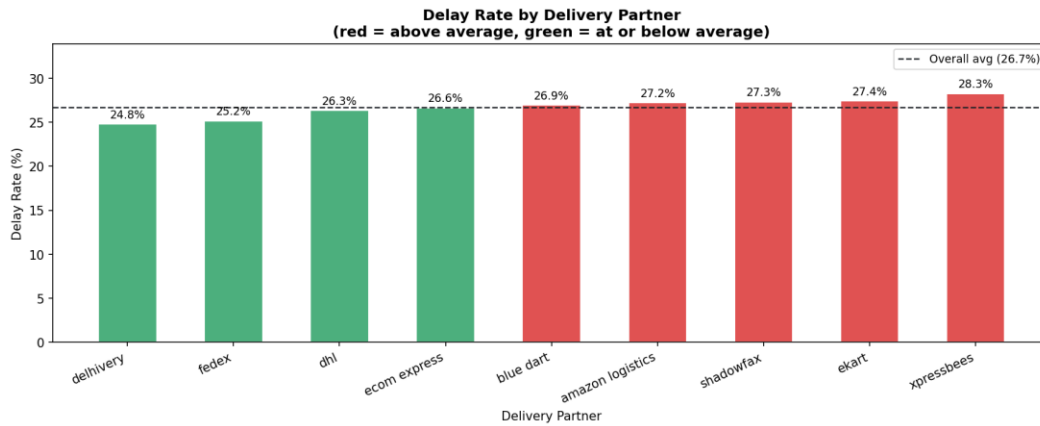


Figure 8: Delay Rate by Delivery Partner (reference line = overall average 26.68%).

15. Region Analysis

Table 10 presents observed delay rates across five geographic regions.

Region	Total	Delayed	Delay Rate	Avg Rating
East	4,919	1,269	25.80%	3.68
North	4,949	1,315	26.57%	3.68
South	4,977	1,333	26.78%	3.67
West	5,095	1,373	26.95%	3.64
Central	5,060	1,379	27.25%	3.66

Table 10: Region analysis (source: region_analysis.csv). Sorted ascending by delay rate. No ranking is implied.

Regional delay rates are very similar across the five regions, ranging from 25.80% to 27.25% — a spread of only 1.45 percentage points. Regional differences may reflect varying mixes of delivery partners, weather conditions, and delivery modes rather than geographic factors alone.

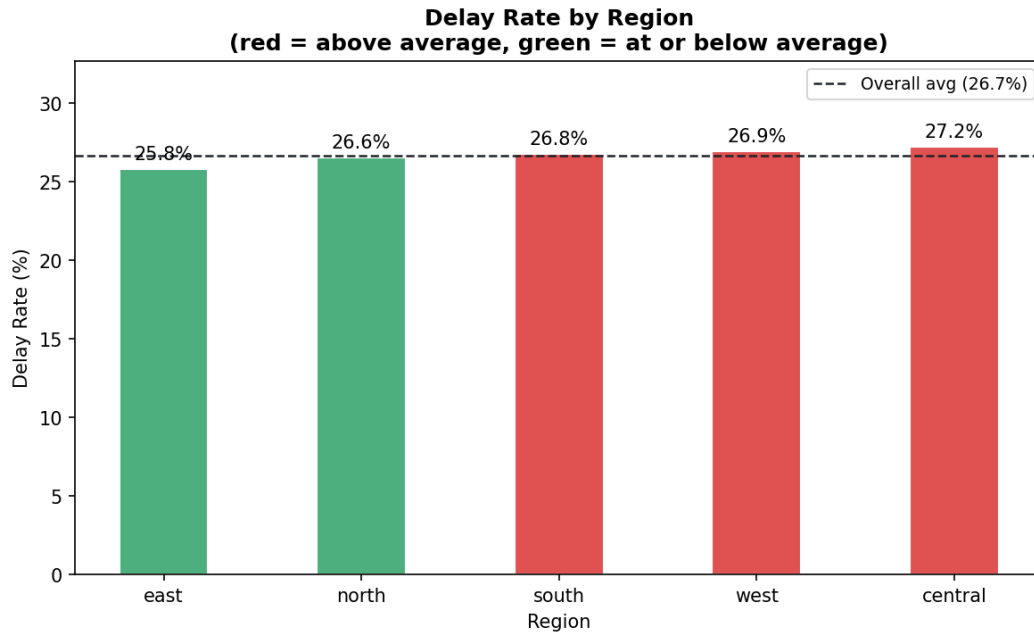


Figure 9: Delay Rate by Region (reference line = overall average 26.68%).

16. Business Insights

The following summarises the major observations from this analysis. All findings are described as patterns observed in this synthetic dataset. Association does not imply causation.

- Overall performance: The dataset records a delay rate of 26.68%, meaning approximately 1 in every 4 deliveries is recorded as delayed.
- Delivery mode: Express mode is associated with the highest observed delay rate (73.78%) and standard mode with the lowest (0%). The 73.78 pp spread is the largest dimensional difference observed in the data. This pattern is partly explained by the much tighter SLA for express deliveries.
- Weather: Stormy conditions are associated with the highest delay rate (41.45%) and cold conditions with the lowest (16.02%). A 25.43 pp spread is observed across weather conditions.
- Distance and delay: Delay rate increases across distance bands, from 17.36% (0–60 km) to 38.28% (241–300 km), a 21 pp increase over the full range.
- Distance and cost: Average delivery cost rises from ₹264.36 to ₹1,463.31 across distance bands — a clear positive association between distance and cost.
- Customer experience: A weak negative correlation ($r = -0.2693$) is observed between delivery time and customer rating. Deliveries rated 1 or 2 all correspond to delayed orders; deliveries rated 4 or 5 show no delays.
- Delivery partners: Partner delay rates span 24.80%–28.27% (range: 3.47 pp). This is a relatively narrow spread. Confounding variables are not controlled for.
- Regions: Regional delay rates span 25.80%–27.25% (range: 1.45 pp), indicating very similar performance across regions in this dataset.

17. Data Quality and Limitations

- Synthetic dataset: The dataset is artificially generated. Results must not be generalised to real-world logistics companies or operations.
- Correlation $\neq$ causation: All relationships described in this report are statistical associations. No factor is claimed to cause delivery delays.
- Zero-hour records: 255 records had an actual delivery time of zero hours. These were flagged and treated as missing actual delivery time in the cleaned working data.
- Duplicate delivery IDs: 500 records are associated with 2 delivery_id values that appear multiple times. These records are preserved in the analysis as a data-quality observation.
- Delay label disagreement: Comparing the original delayed field with a strict 'actual time exceeds expected time' rule produces 1,203 conflicting records. The original delayed field is used throughout as the established source of truth for delay-rate calculations.
- Uncontrolled confounders: Partner and region comparisons do not control for other variables such as delivery mode mix, package types, or weather conditions. Observed differences should be interpreted as descriptive patterns.

18. Conclusion

This project demonstrates a complete Python-based data analytics workflow applied to a synthetic delivery-logistics dataset. The workflow covers data loading, data cleaning, exploratory data analysis, business KPI calculation, dimensional analysis across six key dimensions, visualization generation, and structured findings reporting.

The analysis identifies several notable patterns in the dataset: a 26.68% overall delay rate, wide variation in delay rates across delivery modes, associations between adverse weather and higher delays, a consistent positive relationship between distance and both delivery cost and delay rate, and a weak negative correlation between delivery time and customer satisfaction rating.

The project does not apply machine learning or make predictions about future deliveries. All findings are observations from this synthetic dataset and should not be extrapolated to real-world logistics performance.

19. Future Scope

The following extensions are identified as potential future work and have not been implemented in the current project:

- Apply the same analytical workflow to a validated real-world logistics dataset to test whether the observed patterns hold.
- Build predictive delay models (e.g. classification) using the cleaned dataset if a prediction requirement is introduced.
- Develop interactive dashboards (e.g. using Streamlit or Dash) to allow dynamic exploration of the findings.
- Investigate additional operational factors such as vehicle type, package type, and time of day.
- Extend the partner and region analysis to control for confounding variables using stratification or regression.

20. References and Data Source

Dataset:

- Kaggle – Delivery Logistics Dataset (India – Multi-Partner). Source: Kaggle (exact URL not recorded in project files).

Libraries and documentation:

- Python Software Foundation. Python Programming Language. <https://www.python.org/>
- The Pandas Development Team. pandas: Python Data Analysis Library. <https://pandas.pydata.org/>

- Harris, C.R. et al. Array programming with NumPy. *Nature*, 585, 357–362 (2020). <https://numpy.org/>
- Hunter, J.D. Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90–95 (2007). <https://matplotlib.org/>

Appendix: Project Output Files

All output files are stored in the `business_analysis_outputs/` folder.

A.1 Generated Charts

- `01_kpi_summary.png` — Figure 1: Overall Delivery KPI Summary
- `02_delay_rate_by_mode.png` — Figure 2: Delay Rate by Delivery Mode
- `03_delay_rate_by_weather.png` — Figure 3: Delay Rate by Weather Condition
- `04_distance_vs_delivery_time.png` — Figure 4: Distance Group vs Average Delivery Time
- `05_distance_vs_delivery_cost.png` — Figure 5: Distance Group vs Average Delivery Cost
- `06_delivery_time_vs_rating.png` — Figure 6: Delivery Time vs Customer Rating
- `07_delivery_status_distribution.png` — Figure 7: Delivery Status Distribution
- `08_partner_delay_rate.png` — Figure 8: Delay Rate by Delivery Partner
- `09_region_delay_rate.png` — Figure 9: Delay Rate by Region

A.2 Analysis Result Files

- `kpis.json` — Overall KPI values
- `findings.json` — Structured findings with metrics, interpretations, and cautions
- `mode_analysis.csv` — Delay and time metrics by delivery mode
- `weather_analysis.csv` — Delay and time metrics by weather condition
- `distance_analysis.csv` — Delay, time, and cost metrics by distance band
- `rating_analysis.csv` — Delivery time, cost, and delay rate by customer rating
- `partner_analysis.csv` — Delay rate and average rating by delivery partner
- `region_analysis.csv` — Delay rate and average rating by region

A.3 Project Scripts

- `final_project.py` — Complete analysis workflow (single runnable script)
- `requirements.txt` — Required Python packages
- `README.md` — Project documentation for GitHub
- `process_logistics.py` — Original data-cleaning script
- `eda_logistics.py` — Original EDA script
- `eda_report.py` — Original data-inspection script